

DAMASCUS INTL, Syrian Arab Republic

WMO: 400800

Lat: 33.412N

Lon: 36.516E

Elev: 616

StdP: 94.14

Time Zone: 2.00 (SYR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.2	-12.2	1.4	7.5	-9.8	1.8	9.4	14.6	9.9	12.9	9.8	1.3	350	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	40.0	18.9	38.5	18.5	37.1	18.3	21.5	31.4	20.8	30.4	20.2	29.9	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	15.0	23.6	18.2	14.2	23.2	17.8	13.7	23.0	65.7	31.2	63.2	30.0	60.9	29.9	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	11.0	9.8	-6.1	42.5	2.3	1.6	-7.7	43.6	-9.0	44.6	-10.3	45.4	-12.0	46.6
			-7.3	23.7	2.4	1.6	-9.0	24.8	-10.5	25.7	-11.8	26.6	-13.6	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	7.0	8.9	12.4	16.7	21.7	25.7	28.3	28.1	24.9	19.8	12.9	8.3
	DBStd	8.09	2.56	2.89	3.28	3.61	3.17	2.22	1.71	1.82	2.40	3.29	3.30	2.77
	HDD10.0	237	96	49	12	2	0	0	0	0	0	0	11	67
	HDD18.3	1382	351	265	187	73	8	0	0	0	0	22	165	312
	CDD10.0	3132	4	18	85	203	362	471	566	562	447	304	98	14
	CDD18.3	1237	0	0	2	24	112	221	307	304	198	67	2	0
	CDH23.3	16831	0	4	82	494	1725	3078	4103	3886	2472	933	53	0
	CDH26.7	9727	0	0	20	169	858	1842	2635	2452	1382	363	7	0
Wind	WSAvg	3.9	3.2	3.8	4.2	4.2	4.3	4.9	5.5	4.9	3.9	2.9	2.5	2.8
Precipitation	PrecAvg	172	41	33	23	12	7	0	0	0	0	9	27	41
	PrecMax	279	309	118	135	57	77	15	0	0	7	45	95	303
	PrecMin	57	0	0	0	0	0	0	0	0	0	0	0	1
	PrecStd	58	49	27	24	14	15	2	0	0	1	10	25	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	23.2	29.2	33.7	37.8	40.1	42.6	41.8	39.1	34.9	27.8	20.9
		MCWB	10.3	12.0	13.7	15.4	17.4	18.0	19.0	20.0	18.5	16.9	14.0	11.3
	2%	DB	16.1	19.8	25.7	30.9	35.2	38.2	40.2	39.8	37.0	32.8	24.9	18.0
		MCWB	9.7	10.1	12.5	14.5	16.4	17.6	18.8	19.4	18.2	16.4	13.3	10.7
	5%	DB	14.4	17.4	22.9	28.8	33.8	36.8	38.5	38.1	35.2	30.8	22.8	16.0
		MCWB	8.9	9.4	11.4	13.8	16.0	17.4	18.4	19.0	17.6	16.0	12.6	9.9
	10%	DB	12.9	15.4	20.3	26.1	31.8	35.1	37.0	36.8	33.9	28.7	20.8	14.4
		MCWB	8.2	8.9	10.6	12.8	15.4	17.1	18.4	18.8	17.4	15.5	12.0	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	12.5	15.2	16.9	18.7	20.6	22.2	22.9	21.5	19.8	16.1	14.3
		MCDB	15.3	20.4	24.8	28.7	32.2	33.1	33.1	32.7	30.5	26.9	22.5	16.4
	2%	WB	10.5	11.3	13.4	15.5	17.6	19.5	21.3	21.7	20.2	18.0	14.7	11.9
		MCDB	14.7	18.0	23.0	26.9	31.3	31.6	31.7	31.3	29.2	27.3	20.7	15.5
	5%	WB	9.5	10.2	12.2	14.5	16.8	18.7	20.5	21.0	19.5	17.1	14.0	10.8
		MCDB	13.3	15.8	20.9	26.0	30.0	30.8	30.8	30.3	28.9	26.6	20.0	14.6
	10%	WB	8.8	9.3	11.1	13.5	16.0	18.1	19.9	20.4	18.8	16.3	13.0	10.0
		MCDB	12.3	14.4	18.9	24.5	28.9	30.6	30.2	29.7	28.6	25.8	19.0	13.4
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.8	14.9	16.6	18.2	18.8	18.5	18.4	18.4	17.0	15.1	12.6
		MCDBR	14.3	16.5	18.7	19.9	20.9	20.9	20.6	19.8	20.2	19.3	17.9	15.3
		MCWBR	8.9	8.9	8.7	8.3	7.9	6.7	5.8	5.8	7.1	8.3	9.7	9.7
	5% WB	MCDBR	12.8	13.9	16.7	18.0	18.7	18.9	18.2	18.1	17.9	16.7	14.5	12.6
		MCWBR	8.4	7.9	8.3	7.9	7.3	6.1	5.3	5.5	6.2	7.7	8.2	8.6
Clear-Sky Solar Irradiance	taub	0.340	0.370	0.416	0.467	0.465	0.394	0.398	0.403	0.392	0.415	0.366	0.348	
	taud	2.299	2.197	2.046	1.915	1.956	2.189	2.187	2.170	2.199	2.118	2.244	2.283	
	Ebn at Noon	861	870	854	826	830	889	883	873	868	807	820	826	
	Edh at Noon	104	128	161	192	186	147	146	147	136	136	109	100	
All-Sky Solar Radiation	RadAvg	2.83	3.67	5.17	6.32	7.47	8.40	8.19	7.42	6.29	4.64	3.36	2.63	
	RadStd	0.20	0.29	0.27	0.29	0.34	0.12	0.09	0.17	0.18	0.22	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+102	N/A
(47)	Regional (15 neighbors)	+0.43	N/A	-1.84	N/A	N/A	N/A	N/A	-74	+166	+108

Nomenclature: See separate page